

Eric Huynh

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RESEARCH INTERESTS

How autonomous systems perceive, act, and learn in dynamic environments — spanning learning-based control, reinforcement and imitation learning, multi-sensor fusion, probabilistic state estimation, and field robotics.

EDUCATION

University of Massachusetts Lowell

M.S. in Computer Engineering

Thesis: Sonar-Centric Perception for Underwater Robots: Mapping the Deep, One Angle at a Time (*under preparation*)

Expected: May. 2026

Thesis advisor: Prof. Paul Robinette

University of Massachusetts Lowell

B.S. Double Major: Electrical Engineering and Computer Science, Minor in Robotics

Sept. 2020 — Dec. 2023

GPA: 3.94/4.00

Massachusetts Bay Community College

A.S. Electrical and Computer Engineering

Sept. 2018 — May 2020

GPA: 3.985/4.00

PUBLICATIONS

Conference publications (peer-reviewed)

2. **E. Huynh**, M. E. Cabrera, P. Robinette, “Spectral Whitening and Confidence Fusion for Robust Sonar Scan Matching,” 2026 IEEE International Conference on Robotics and Automation (ICRA). *Under Review*
1. **E. Huynh** and P. Robinette, “Modality Matters: A Sim-to-Real Study of Sonar-Based Object Detection and Tracking,” OCEANS 2025 - Great Lakes, Chicago, IL, USA, 2025, pp. 1-6, doi: 10.23919/OCEANS59106.2025.11244929.

Workshop Papers and Posters (lightly peer-reviewed)

4. **E. Huynh** and P. Robinette, “Sim-to-Real Evaluation of Scanning and Multibeam Sonar for Object Detection in Underwater Environments,” 2025 ICRA Workshop AQ²UASIM: Advancing Quantitative and QUALitative SIMulators for marine applications, Atlanta, GA, USA, May 2025.
3. **E. Huynh** and P. Robinette, “Evaluating Sonar-Based Object Detection in HoloOcean And Real-World Environments,” 2025 UMass Lowell Student Symposium, April 2025.
2. **E. Huynh**, S. Sicari and P. Robinette, “Optimizing Underwater Image Enhancement Algorithms Through HoloOcean Simulator,” 2024 Northeast Robotics Colloquium (NERC), Amherst, MA, USA, September 2024.
1. **E. Huynh**, N. Uhunsere, Z. R. Khavas and P. Robinette, “A 3D Simulation Approach to Evaluate Human-Robot Interaction in Emergency Scenarios,” 2024 UMass Lowell Student Symposium, April 2024.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant, UMass Lowell NERVE Center

University of Massachusetts Lowell

June 2024 — Present

Lowell, MA

- Develop and evaluate sonar-centric SLAM and multi-sensor fusion algorithms (sonar, IMU, camera, USBL) for underwater perception, mapping, and localization on simulated and real BlueROV2 platforms, with results presented at OCEANS 2025 and workshop at ICRA 2025.

Research Assistant, US DEVCOM ARL STRONG

University of Massachusetts Lowell

July 2022 — May 2024

Lowell, MA

- Co-designed a 3D Unreal Engine simulation to study and improve human-robot interaction and task engagement in collaborative emergency situations for robotics research.

Reliability Operations Engineering Intern, Reliability Operations Team Jan. 2022 — June 2022
Analog Devices Wilmington, MA

- Redesigned LabVIEW-based control software and developed automated monitoring tools to enhance hardware integration, reliability testing, and data collection efficiency in the Reliability Operations Lab.

Undergraduate Research Assistant May 2021 — Dec. 2021
University of Massachusetts Lowell Lowell, MA

- Assisted Professor Noah Jafferis in translating his doctoral research codebase from LabVIEW to MATLAB.

Oculus Launch Pad Fellow Sep. 2020 — April. 2021
Oculus VR Menlo Park, CA

- Developed a VR prototype using Unity and Oculus SDK as part of the Oculus Launch Pad Fellowship, focusing on immersive interaction design and performance optimization.

Research Assistant, SiliconSynapse Lab April 2020 — Sep. 2020
Northeastern University Boston, MA

- Assisted in the design and development of HARPY, a thruster-assisted legged robot prototype.

Research Volunteer May 2019 — May 2020
Massachusetts Bay Community College Wellesley, MA

- Supported the integration of VR/AR applications at MassBay to prepare students for the workforce and gain hands-on experience.

Peer Tutor Feb 2019 — May 2020
Massachusetts Bay Community College Wellesley, MA

- Peer Tutor in MATLAB and Mathematics, including Algebra (all levels), Calculus I–III, and Differential Equations.

GRANTS AND FELLOWSHIPS

UMass Lowell International Student Scholarship	2020 — 2023
Debi Prasad Sodhani Scholarship	Jan. 2020
MassBay STEM Mentor Program Scholarship	Jan. 2020

HONORS AND AWARDS

UMass Lowell Dean's Medal for Highest Achievement in Electrical Engineering	May 2021
Semi-finalist, UMass Lowell DifferenceMaker Idea Challenge — campus-wide innovation competition	May 2021
MassBay IMPACT Award	May 2020
MassBay Electrical and Computer Engineering Program Award Winner	May 2020

PROFESSIONAL SERVICE

Reviewer

- IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR) Reviewer Aug. 2024

Leadership

- MassBay Engineering Club Co-President 2019 — 2020

Volunteer Service & Outreach

- Mentoring and Academic Support (MAS) Program at Walsh Middle School 2018 — 2019